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1 - OVERVIEW

This section applies to *TwinSpeed* Magnet systems and explains the procedure for the replacement of the Vacuum pump.

Parts mentioned are listed in Table 1-1:

TABLE 1-1
FRU(1) PART LIST

Manufacturer's Part Number	GEMS Part Number	Item Description
	2282045	Vacuum Pump

Special tools needed to replace the Vacuum Pump are listed in Table 1-2:

TABLE 1-2
SPECIAL TOOLS NEEDED

Item	Description	GE Part Number
1	Universal Lift Hoist Kit with TAC attachments	46-317266G5
2	Small hand tools including a Phillips-head screwdriver	
3	Two medium-sized crescent wrenches	

2 - VACUUM PUMP REMOVAL

DANGER!!

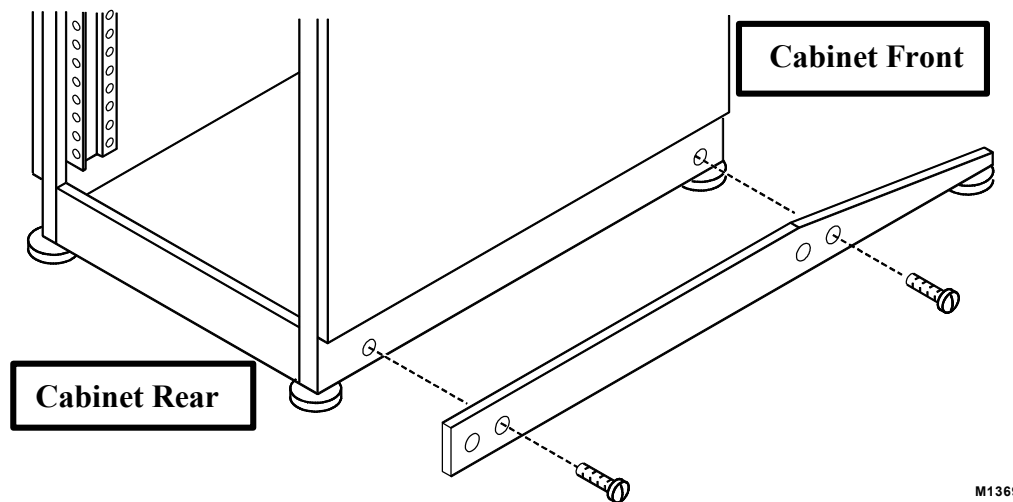
POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE TAC CABINET BEFORE ATTEMPTING TO REMOVE THE VACUUM PUMP. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

1. Lock out and tag out the TAC cabinet. (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

DANGER!!

CABINET TIP HAZARD! POSSIBLE PERSONAL INJURY! CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE AN RF AMPLIFIER.

2. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the TAC Cabinet. See Illustration 2-1.



INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET
ILLUSTRATION 2-1

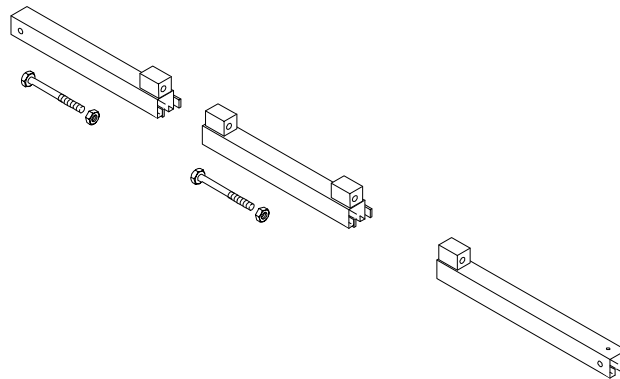
3. Remove the front cover and the top bracket from the TAC cabinet.
4. Open the rear TAC cabinet door.
5. Remove the four (4) screws that attach the fan module to the TAC cabinet. Set the fan module in the cabinet.
6. Unplug the Vacuum Pump power cord from the side of the vacuum pump.
7. Remove the vacuum connector at the vacuum pump.

8. Locate and open the Universal Lift Hoist Kit.

NOTE

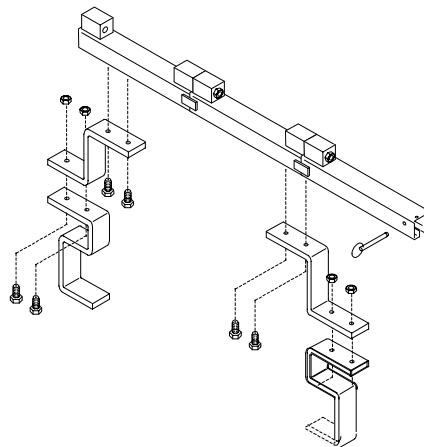
The Universal Lift Hoist is assembled from parts stored in the Universal Lift Hoist Kit. Lift attachments specially designed for the vacuum pump are located in the cardboard box at the bottom of the TAC Cabinet.

9. Locate the three (3) large Steel Rail sections and place them on a flat surface for assembly. See Illustration 2-2.



STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS
ILLUSTRATION 2-2

10. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75" X 1/2" caps crews and hex nuts.
11. Use the crescent wrenches to tighten the cap screws.
12. Assemble and fasten the elevating brackets to the rail assembly. See Illustration 2-3.



HOIST TOP RAIL PARTS WITH ELEVATING BRACKETS
ILLUSTRATION 2-3

13. Center the Steel Rail and elevating bracket assembly at the top of TAC Cabinet, so the hoist assembly will face the front of the cabinet. See Illustration 2-4.



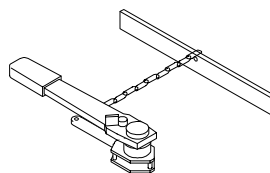
HOIST INSTALLED ON TAC CABINET
ILLUSTRATION 2-4

14. Tighten the bolts on the Steel Rail to secure it to the top of the cabinet.



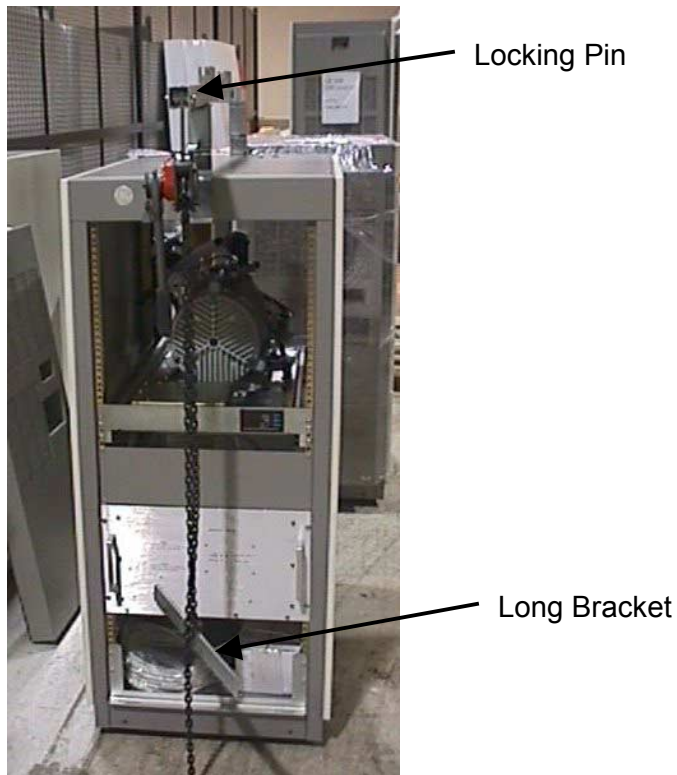
VERIFY THAT ALL HARDWARE IS SECURELY FASTENED; ALL BOLTS AND NUTS MUST BE TIGHT PRIOR TO ATTACHING THE HOIST ASSEMBLY TO THE LIFT KIT.

15. Locate the Hoist Assembly in the Lift Kit. See Illustration 2-5.



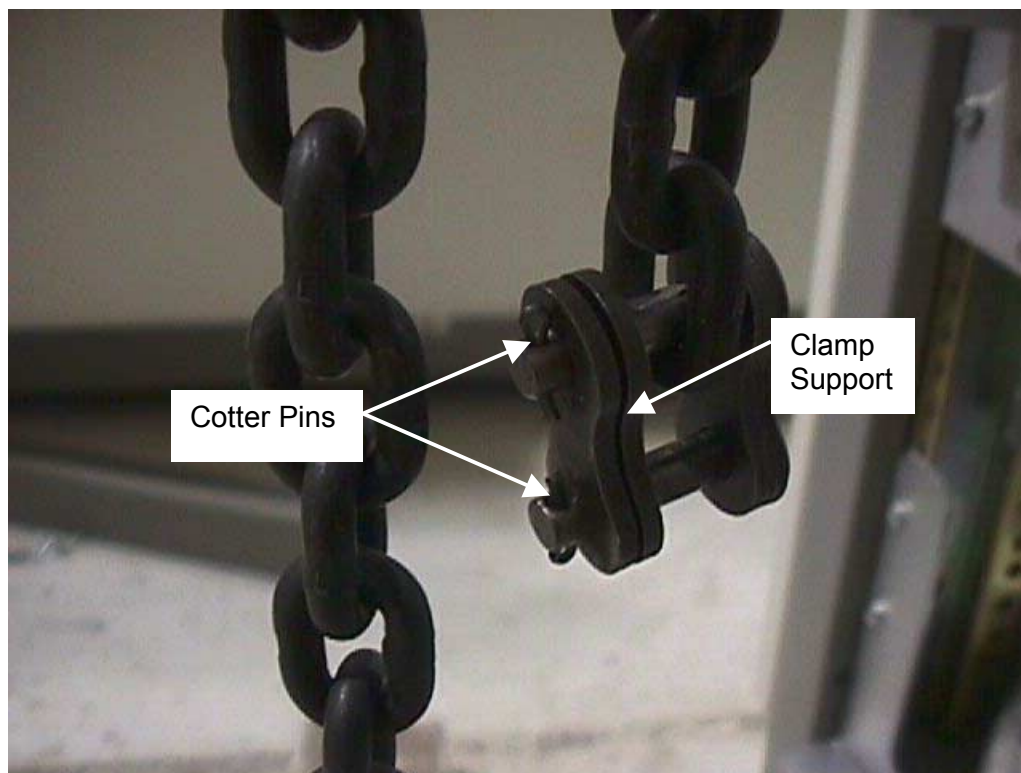
HOIST ASSEMBLY
ILLUSTRATION 2-5

16. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See Illustration 2-6.



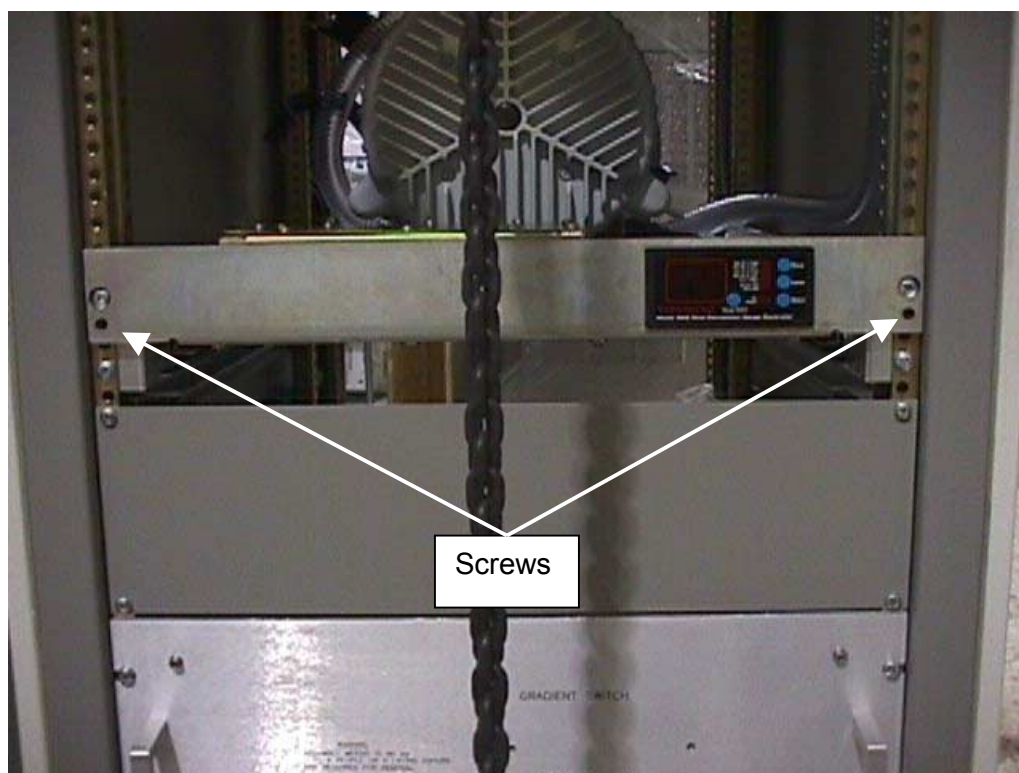
PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY
ILLUSTRATION 2-6

17. Remove the two cotter pins that hold the clamp to the hoist chain. See Illustration 2-7.



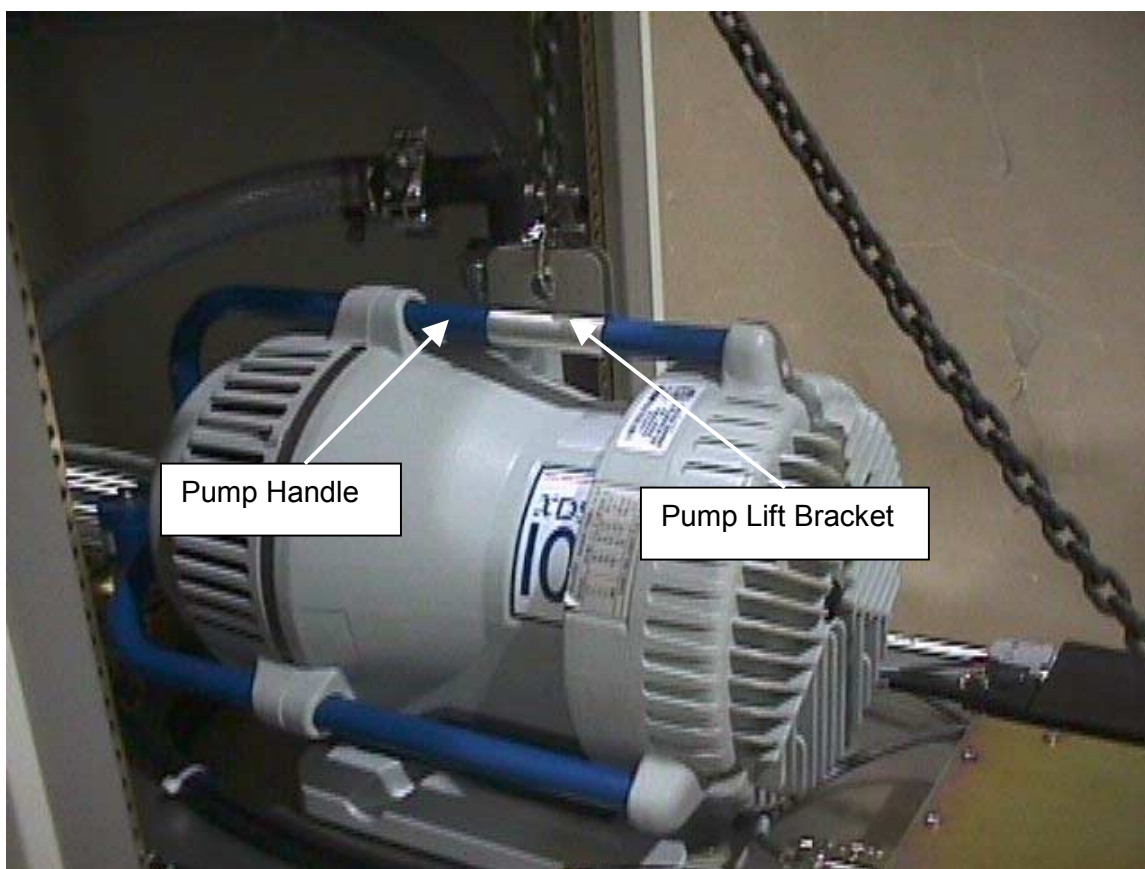
HOIST CHAIN WITHOUT A BRACKET
ILLUSTRATION 2-7

18. Remove the clamp support.
19. Remove the long bracket. See Illustration 2-6.
20. Replace the clamp support and cotter pins.
21. Remove the two screws that hold the vacuum pump shelf in place. See Illustration 2-8.



VACUUM PUMP SHELF
ILLUSTRATION 2-8

22. Slide the vacuum pump out of the cabinet until you can access the pump handle.
23. Attach the vacuum pump lift bracket to the pump handle. The vacuum pump lift bracket is located in the cardboard box at the bottom of the TAC cabinet. See Illustration 2-9.
24. Attach the vacuum pump lift bracket to the hoist.



VACUUM PUMP LIFT BRACKET
ILLUSTRATION 2-9

25. Remove the four (4) bolts that connect the vacuum pump to the shelf.
26. Hoist the vacuum pump up and out of the cabinet, then place it safely on the floor.
27. Remove the vacuum pump lift bracket from the vacuum pump.

3 - VACUUM PUMP INSTALLATION

1. Attach the vacuum pump lift bracket to the pump handle. See Illustration 2-9.
2. Hoist the vacuum pump up and place is safely in place on the pump shelf.
3. Install the four (4) bolts that attach the vacuum pump to the shelf.
4. Remove the vacuum pump lift bracket from the vacuum pump.
5. Slide the vacuum pump shelf back into place.
6. Install the two screws that hold the vacuum pump shelf in place. See Illustration 2-8.
7. Remove the hoist assembly from front rail. See Illustration 2-6.
8. Loosen the bolts that secure the Steel Rail to the top of the cabinet.
9. Remove the Steel Rail from the cabinet.
10. Reattach the vacuum connector to the vacuum pump.
11. Plug the vacuum pump power cord into the side of the vacuum pump.
12. Replace the top bracket and front cover for the TAC Cabinet.
13. Replace the fan module and secure with four (4) screws.
14. Close the rear TAC cabinet door.
15. Remove the anti-tip legs. See Illustration 2-1.
16. Remove the lock out and tag out for the TAC Cabinet.
17. Un-assemble and return the Universal Lift Hoist Kit components to the case and the vacuum pump lift bracket to the TAC Cabinet.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Oct. 24, 2001	K. Schraufnagel	Preliminary release